

Securing MySQL Server

The MySQL server is widely used in the open source world, whether it's for the financial data of SMBs or mission-critical ERP systems. This is due to its great performance and compatibility. However, it is also plagued with security challenges, just like any open source database solution. Utmost care needs to be taken to configure MySQL in order to strengthen its security. This article talks about the challenges and solutions.



The MySQL architecture is based on a typical database engine model. It consists of a storage manager, query processing engine, the authentication engine and client connectors at its core. As for the database engine, we all know that databases are a structured collection of data in a digital form. The structure consists of schema definitions; tables, which store the data in an organised form; views, which are virtual tables; and stored queries, to improve performance, etc. The database engine accepts SQL query requests, executes them and returns data to the querying computer. For this purpose, the database service listens on one or more TCP/IP ports. The service also runs closely tied with the operating system, to gain better control on disk and memory resources. As for authentication, MySQL allows you to create users and roles, whereby each user logging into the database is bound by a pre-defined security context.

Each of the MySQL components mentioned above is vulnerable to some extent, if not configured properly. We must remember that the database is a widely and continuously accessible component, which makes it more vulnerable and susceptible to attacks. MySQL security requirements touch all networking layers, and hence need careful design and implementation. While the MySQL guide is a great online resource, it only talks about securing the database layer. Listed below are various steps that need to be taken after installing a MySQL server.

Securing MySQL, post-installation

Harden the operating system: The correct way to secure a database server starts with hardening the OS on which it is installed. The security documentation for your distro can be followed to ensure a healthy OS installation, but do ensure that the following important aspects are taken care of:

- Set up an iptables firewall.
- Disable unnecessary services.
- Ensure a strong password policy.
- Install anti-virus and anti-spam software.
- Update to the latest security patch level.

Set **chroot**: Once installed, the MySQL service should be configured to be **chrooted**. This is essential to restrict the MySQL service's jurisdiction in terms of mounted disk volume usage, and to ensure privilege separation for database applications and the query engine. The command `mkdir -p /chroot/mysql/etc` can be used to set up a **chrooted** /etc folder, and can be repeated for other important folders such as /dev, /var/tmp, etc. Once created, the **chown** command can be used to set access rights, so that the MySQL daemon has control over them.

Lower the service privileges: It is not necessary to run the MySQL service with root-level privileges. This can create security challenges, whereby the attacker can run malicious code in the same security context as MySQL, thus gaining complete